HR ANALYTICS DASHBOARD

Employee Attrition, Workforce, Experience & Performance Analysis

Project Report

Tool: Microsoft Excel + Microsoft Power BI

Prepared from the submitted Power BI (.pbix) project

Project Information

Item	Details
Project Title	HR Analytics Dashboard
Organization / Case	XYZ Corporation
Primary Tool	Microsoft Power BI
Data Preparation	Microsoft Excel / Power Query
Analysis Areas	Employee Overview, Attrition, Workforce, Experience & Performance
Dashboard Pages	Home, Attrition Analysis, Workforce Analysis, Experience & Performance Analysis

Table of Contents

1. 1. Abstract
2. 2. Introduction
3. 3. Problem Statement
4. 4. Project Objectives
5. 5. Scope of the Project
6. 6. Dataset and Excel Data Cleaning
7. 7. Data Preparation and Transformation
8. 8. Power BI Data Model and DAX
9. 9. Dashboard Design and Navigation
10. 10. Page 1 – HR Dashboard Home
11. 11. Page 2 – Attrition Analysis
12. 12. Page 3 – Workforce Analysis
13. 13. Page 4 – Experience & Performance Analysis
14. 14. Slicers and Interactivity
15. 15. Key Business Questions Answered
16. 16. Business Insights – How to Derive Them
17. 17. Recommendations
18. 18. Testing and Validation
19. 19. Limitations
20. 20. Conclusion
21. 21. Future Scope

1. Abstract

The HR Analytics Dashboard is an interactive business intelligence project designed to help HR teams understand employee attrition, workforce composition, and employee experience. The project uses employee-level HR data, prepares the data for analysis, creates measures and calculated columns in Power BI, and presents the results through interactive dashboard pages. The dashboard focuses on employee totals, employees who left, attrition rate, age, salary, department, job role, education, gender, experience, job satisfaction, work-life balance and performance rating.

The submitted Power BI file contains a dedicated home/navigation page and three analysis pages: Attrition Analysis, Workforce Analysis, and Experience & Performance Analysis. The dashboard uses KPI cards, slicers, bar/column charts, line charts, pie/donut-style visuals and navigation buttons to support interactive exploration.

2. Introduction

Human Resource departments generate large amounts of employee data, but raw tables do not automatically provide useful business information. HR analytics converts employee records into measurable indicators that can be used to understand workforce structure and employee exits.

For this project, the main business problem is to understand employee attrition and identify patterns associated with department, role, salary, age, experience, education, overtime, gender, marital status, job satisfaction, work-life balance and performance.

3. Problem Statement

XYZ Corporation has more than 1,400 employees across major business functions such as HR, Research & Development and Sales. The HR team needs an interactive way to monitor workforce size, employee exits and factors associated with attrition instead of relying only on raw Excel records.

Business Questions

- How many employees are currently represented in the dataset?
- How many employees have left the organization?
- What is the attrition rate?
- What are the average age, salary and years at company?
- Which departments and job roles have higher attrition counts?
- How does attrition vary by age, education, overtime, salary, gender and marital status?
- How is the workforce distributed across department, education, gender, salary and experience?
- How does attrition vary with years at company, total working years and years in the current role?
- How do job satisfaction and work-life balance relate to employee attrition?
- How does performance rating relate to employee retention?

4. Project Objectives

22. Clean and prepare the HR employee data before analysis.
23. Load the prepared data into Power BI.
24. Create DAX measures for important HR KPIs.
25. Create useful calculated columns for age, salary and experience grouping.

26. Build an interactive HR dashboard with slicers and navigation.
27. Analyze attrition across employee and job-related dimensions.
28. Analyze workforce composition and experience/performance.
29. Present findings in a form that is easy for HR stakeholders to understand.

5. Scope of the Project

- Employee overview and KPI monitoring.
- Attrition analysis by demographic, job and compensation-related dimensions.
- Workforce distribution analysis.
- Experience and performance analysis.
- Interactive filtering through slicers.
- Navigation between dashboard pages.
- DAX-based measures and calculated columns.
- Excel/Power Query data-cleaning workflow before visualization.

6. Dataset and Excel Data Cleaning

Before building the dashboard, the raw employee data should be checked and cleaned. The cleaning stage is important because incorrect data types, duplicate employee records, blank values or inconsistent category labels can produce incorrect Power BI results.

6.1 Recommended Excel Data-Cleaning Process

Step	Cleaning Activity	Purpose
1	Backup raw data	Keep an untouched copy of the original Excel file before making changes.
2	Check column names	Use clear, consistent names such as Attrition, Department, JobRole, Age and MonthlyIncome.
3	Remove duplicate records	Check for duplicate employee records and remove only true duplicates.
4	Handle blank values	Find blanks in important fields and decide whether to correct, replace or exclude them.
5	Clean text	Use TRIM/CLEAN where needed and standardize spelling/case of category values.
6	Set data types	Age, years and ratings should be numeric; Attrition, Department and JobRole should be text.
7	Validate numeric ranges	Check for impossible or unexpected values in age, salary, ratings and experience fields.
8	Check category consistency	Make sure categories such as Yes/No and department names use one standard representation.
9	Check salary values	Confirm MonthlyIncome is numeric and contains no currency symbols stored as text.
10	Save cleaned dataset	Keep a clean version for Power BI import and retain the raw version separately.

6.2 Useful Excel Cleaning Functions

Function	Example	Use
TRIM	=TRIM(A2)	Removes unnecessary spaces.

CLEAN	=CLEAN(A2)	Removes non-printing characters.
IFERROR	=IFERROR(A2,"")	Handles formula errors.
COUNTIF	=COUNTIF(A:A,A2)	Helps identify repeated values.
Data Validation	Yes / No list	Keeps categorical values consistent.
Remove Duplicates	Data → Remove Duplicates	Removes exact duplicate rows.

Important: the cleaning steps should be applied according to the actual raw file. If a field is already clean, it should not be unnecessarily changed.

6.3 Power Query Cleaning Workflow

30. Open Power BI and choose Get Data → Excel.
31. Select the HR data worksheet/table.
32. Choose Transform Data to open Power Query.
33. Check headers and column names.
34. Set the correct data type for each column.
35. Remove duplicates if duplicate employee records are confirmed.
36. Trim and clean text fields where necessary.
37. Replace inconsistent category values where verified.
38. Check null/blank values in important columns.
39. Close & Apply after validation.

7. Data Preparation and Transformation

The dashboard uses the HR_Analytics table in Power BI. The submitted PBIX also contains calculated fields and measures used by the report.

7.1 Calculated Columns Used

Age Group:

Age Group = SWITCH(TRUE(), [Age] <= 25, "18-25", [Age] <= 35, "26-35", [Age] <= 45, "36-45", [Age] <= 55, "46-55", "56+")

Salary Group:

Salary Group = SWITCH(TRUE(), [MonthlyIncome] < 3000, "Below 3K", [MonthlyIncome] < 6000, "3K - 6K", [MonthlyIncome] < 10000, "6K - 10K", [MonthlyIncome] < 15000, "10K - 15K", "15K+")

Experience Group:

Experience Group = SWITCH(TRUE(), [TotalWorkingYears] <= 2, "0-2 Years", [TotalWorkingYears] <= 5, "3-5 Years", [TotalWorkingYears] <= 10, "6-10 Years", [TotalWorkingYears] <= 20, "11-20 Years", "20+ Years")

8. Power BI Data Model and DAX

The project uses DAX measures to calculate KPIs dynamically. Measures respond to slicers and visual filters, which makes the dashboard interactive.

Measure	DAX Logic	Business Meaning
Total Employees	COUNTROWS(HR_Analytics)	Total employee records
Attrition Count	CALCULATE(COUNTROWS(HR_Analytics), HR_Analytics[Attrition] = "Yes")	Employees who left

Active Employees	CALCULATE(COUNTROWS(HR_Analytics), HR_Analytics[Attrition] = "No")	Employees who stayed
Attrition Rate	DIVIDE([Attrition Count], [Total Employees], 0)	Attrition percentage
Average Age	AVERAGE(HR_Analytics[Age])	Average employee age
Average Monthly Salary	AVERAGE(HR_Analytics[MonthlyIncome])	Average monthly income
Average Years At Company	AVERAGE(HR_Analytics[YearsAtCompany])	Average tenure

9. Dashboard Design and Navigation

The submitted PBIX contains four pages in the following order. The home page is intended as a navigation screen, while the remaining three pages provide the analysis.

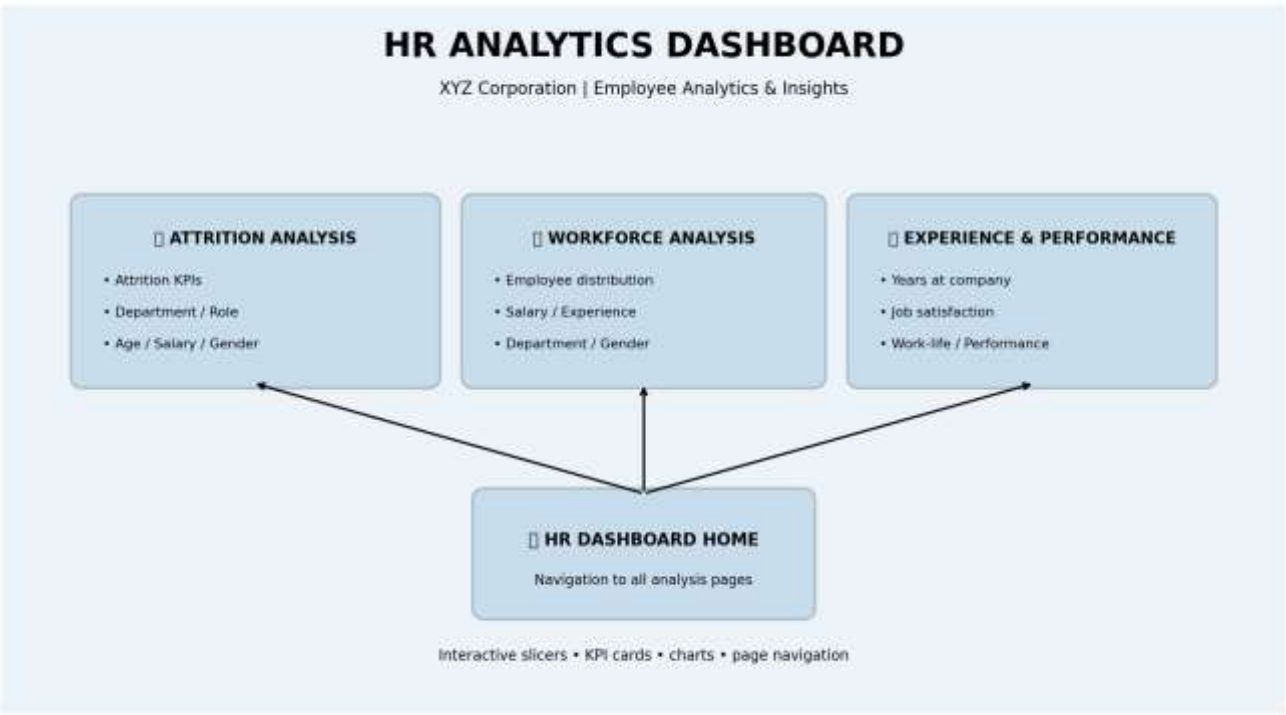


Figure 1. Dashboard structure and navigation used in the submitted report.

Page	Purpose	Main Elements
HR Dashboard Home	Landing/navigation page	Title, three navigation buttons, analysis selection
Attrition Analysis	Understand employee exits	KPI cards, salary/department/age/gender visuals, slicers, navigation
Workforce Analysis	Understand current workforce composition	Employee totals, average salary/age/experience, department/gender/salary/experience visuals, slicers
Experience & Performance Analysis	Study tenure and employee experience	Years at company, total experience, current role, job satisfaction, work-life balance, performance rating, slicers

10. Page 1 – HR Dashboard Home

The Home page is a navigation page. It provides a clear starting point for the report and lets the viewer move directly to the three analytical pages.

- HR Analytics Dashboard title.
- Employee Analytics & Insights subtitle.
- Attrition Analysis navigation button.
- Workforce Analysis navigation button.
- Experience & Performance navigation button.
- Consistent blue technology-style background.

11. Page 2 – Attrition Analysis

This page focuses on employees who have left the organization and supports investigation of attrition across salary, department, age and other filters.

Component	Purpose
Total employee KPI	Shows the overall employee count in the current filter context.
Attrition count KPI	Shows the number of employees with Attrition = Yes.
Attrition rate KPI	Shows attrition as a proportion of total employees.
Average age KPI	Shows average employee age.
Average years at company KPI	Shows average tenure.
Salary-based visual	Shows attrition/workforce distribution across salary groups.
Department visual	Compares attrition across departments.
Age visual	Shows attrition across age groups.
Gender visual	Supports demographic attrition analysis.
Slicers	Allow the user to filter the page interactively.

12. Page 3 – Workforce Analysis

The Workforce Analysis page focuses on the composition of the employee population rather than only on who left. It provides a view of department, gender, salary and experience distribution.

Component	Purpose
Total employee KPI	Shows the current employee count.
Average salary KPI	Shows average salary/income.
Average age KPI	Shows average age.
Average experience KPI	Shows average experience/tenure indicator used on the page.
Department distribution	Shows how employees are distributed by department.
Gender distribution	Shows workforce composition by gender.
Salary distribution	Shows employees across salary groups.
Experience distribution	Shows employees across experience groups.
Slicers	Allow interactive workforce filtering.

13. Page 4 – Experience & Performance Analysis

This page analyzes the relationship between employee tenure/experience and employee-related experience or performance indicators. The project brief specifically includes years at company, total working years, years in current role, job satisfaction, work-life balance and performance rating.

Analysis	Visual / Field	Purpose
Attrition by Years at Company	Line chart; YearsAtCompany + Attrition Count	Identify attrition variation by tenure.
Attrition by Total Working Experience	Column chart; Experience Group + Attrition Count	Compare attrition across experience bands.
Attrition by Years in Current Role	Column/bar chart; YearsInCurrentRole + Attrition Count	Study exits by current-role tenure.
Attrition by Job Satisfaction	Column chart; Job Satisfaction + Attrition Count	Compare exits across satisfaction levels.

Attrition by Work-Life Balance	Column chart; WorkLifeBalance + Attrition Count	Compare exits across work-life balance levels.
Performance Rating vs Retention	PerformanceRating with attrition/active employees	Compare retention and attrition by performance rating.

14. Slicers and Interactivity

Slicers allow users to interactively filter dashboard visuals. The submitted report includes slicers on the analytical pages, including categorical and range-based filtering.

- Department – filter analysis to a specific department.
- Gender – compare workforce or attrition for selected gender categories.
- Age / Age Group – analyze a selected age range or age category.
- Job Role – focus on a particular role.
- Salary or other range filters – examine a selected numeric range where configured.

When a slicer is selected, measures such as Total Employees, Attrition Count and Attrition Rate recalculate in the current filter context.

15. Key Business Questions Answered

Analysis Area	Questions Supported
Employee overview	Total employees, employees who left, attrition rate, average age, salary and years at company.
Attrition	Attrition by department, job role, age, education, overtime, salary, gender and marital status where those fields are used in the analysis.
Workforce	Distribution of employees by department, education, gender, salary and experience.
Experience & performance	Attrition by years at company, total working years, years in current role, job satisfaction, work-life balance and performance rating.

16. Business Insights – How to Derive Them

The final insights should be written only after reading the actual values from the dashboard. The following are safe insight templates; they avoid inventing results that are not verified from the data.

40. The department with the highest displayed attrition count is _____, based on the Attrition by Department visual.
41. The age group with the highest displayed attrition count is _____.
42. The salary group with the highest displayed attrition count is _____.
43. The experience group with the highest displayed attrition count is _____.
44. Attrition varies across job satisfaction levels; the highest displayed category is _____.
45. Attrition varies across work-life balance levels; the highest displayed category is _____.
46. The performance rating category with the highest displayed employee exit count is _____.
47. The workforce is concentrated in _____ department based on the employee distribution visual.
48. The workforce salary distribution is concentrated in _____ salary group.
49. The workforce experience distribution is concentrated in _____ experience group.

17. Recommendations

Recommendations should be connected to the actual patterns observed in the dashboard. The following are decision areas that HR can investigate when the corresponding visual shows a meaningful pattern.

- Investigate departments or job roles showing comparatively high attrition counts.
- Review employee satisfaction and work-life balance when those categories show elevated exits.
- Review compensation structure when salary-related attrition patterns are visible.
- Examine early-tenure attrition and onboarding if lower-tenure groups show higher exits.
- Review career progression and role mobility when attrition is concentrated around current-role tenure.
- Use performance and retention analysis together rather than treating performance rating alone as a cause of attrition.
- Use slicers to drill down before taking HR action; aggregate patterns should be validated with HR context.

18. Testing and Validation

Test	Expected Result
Total Employees	Matches the number of valid employee records represented in the model.
Attrition Count	Counts only records where Attrition = Yes.
Active Employees	Counts only records where Attrition = No.
Attrition Rate	Attrition Count divided by Total Employees.
Slicer interaction	Charts and KPI cards respond to selected filters.
Page navigation	Home and navigation buttons open the intended report page.
Data types	Numeric fields behave numerically and categorical fields behave as categories.
Category sorting	Grouped categories appear in logical business order where sort columns are configured.

19. Limitations

- The dashboard describes patterns in the available employee dataset; it does not by itself prove causation.
- Attrition counts can be affected by the number of employees in each category, so counts should be interpreted with the relevant workforce size.
- The dashboard does not replace HR interviews, employee feedback or organizational context.
- Any missing, inconsistent or incorrectly typed source data can affect results.
- External factors such as market conditions or policy changes are outside the dataset unless explicitly included.

20. Conclusion

The HR Analytics Dashboard converts employee-level HR data into an interactive Power BI report. The workflow covers Excel/Power Query data cleaning, data preparation, DAX-based KPI creation, interactive slicers, dashboard navigation and multi-dimensional HR analysis.

The final report contains a Home page for navigation and three analytical pages covering attrition, workforce distribution, and experience/performance. This structure provides a practical foundation for HR teams to explore employee patterns and identify areas that require further investigation.

21. Future Scope

- Add a dedicated employee-level drill-through page.
- Add time-based attrition trends if historical employee snapshots become available.
- Add predictive attrition modeling using a suitable machine-learning workflow.
- Add HR cost impact of attrition where salary and replacement-cost data are available.
- Add automated refresh from a governed HR data source.
- Add role-based security if the dashboard is deployed to different HR users.

Appendix A – Suggested Final Dashboard Checklist

- ☐ Home page is first and contains navigation to all analysis pages.
- ☐ Attrition Analysis page contains KPI cards, visuals and slicers.
- ☐ Workforce Analysis page contains workforce distribution visuals.
- ☐ Experience & Performance page contains all six required analysis areas.
- ☐ Grouped categories are logically sorted (for example, 0–2, 3–5, 6–10, 11–20, 20+).
- ☐ Job Satisfaction and Work-Life Balance categories are ordered Low → Medium → High → Very High.
- ☐ Performance Rating visual clearly distinguishes attrition from active/retained employees where used.
- ☐ Titles and fonts are consistent.
- ☐ Background is readable and does not reduce chart readability.
- ☐ Navigation buttons work in Power BI Reading/Presentation mode.
- ☐ Slicers affect the intended visuals.
- ☐ Final business insights are based on the actual displayed data.

Source note: This report is based on the submitted Power BI project structure and the HR Analytics project brief. Where exact dashboard result values are not explicitly verified in the source, the report uses descriptions or placeholders rather than inventing findings.